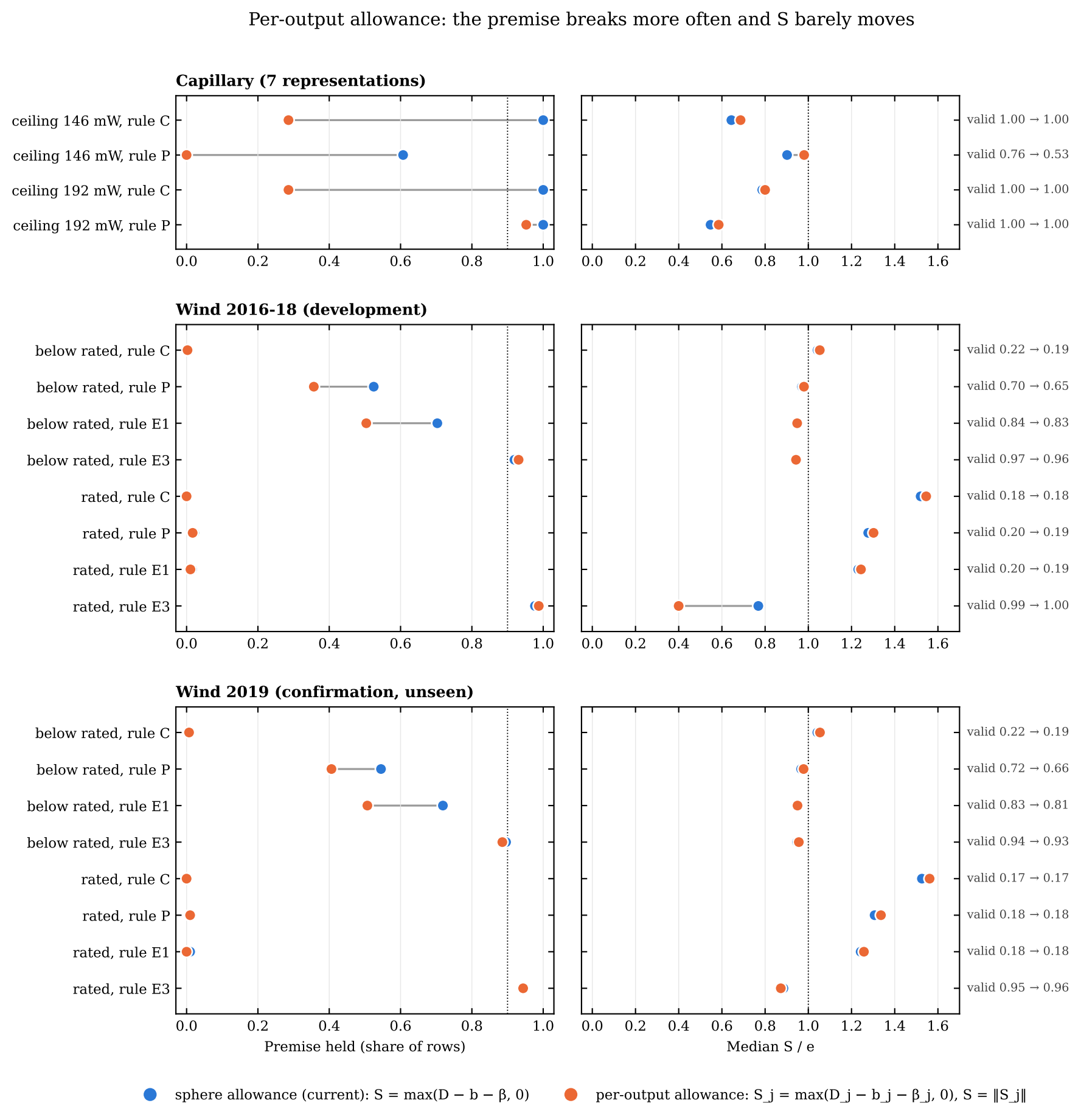
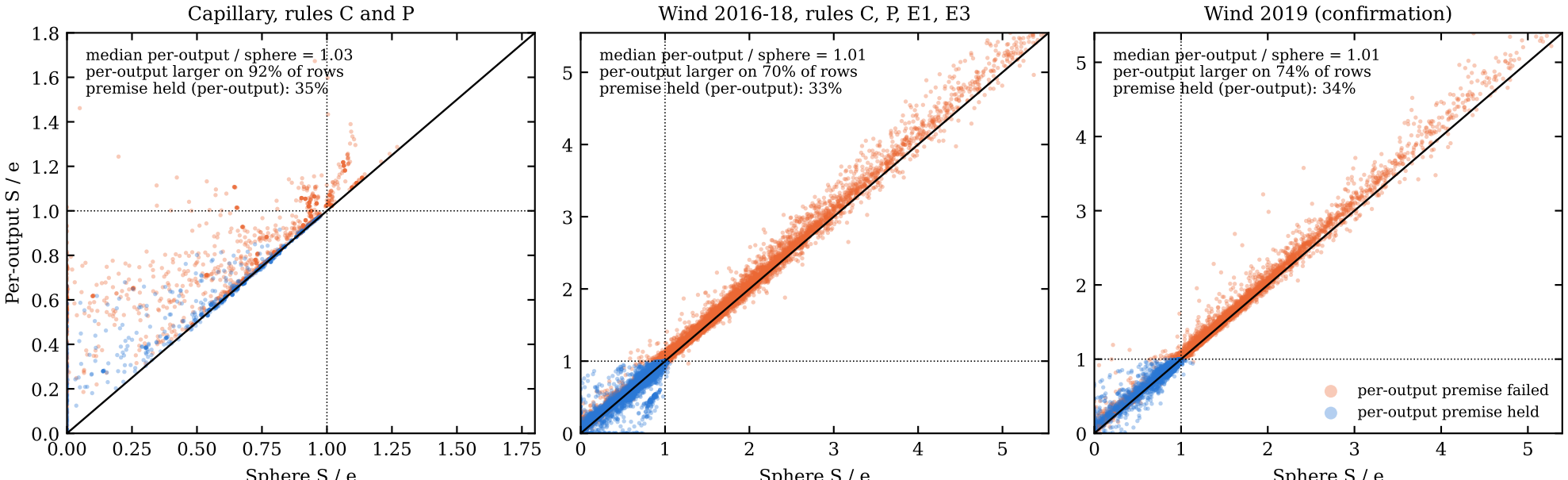
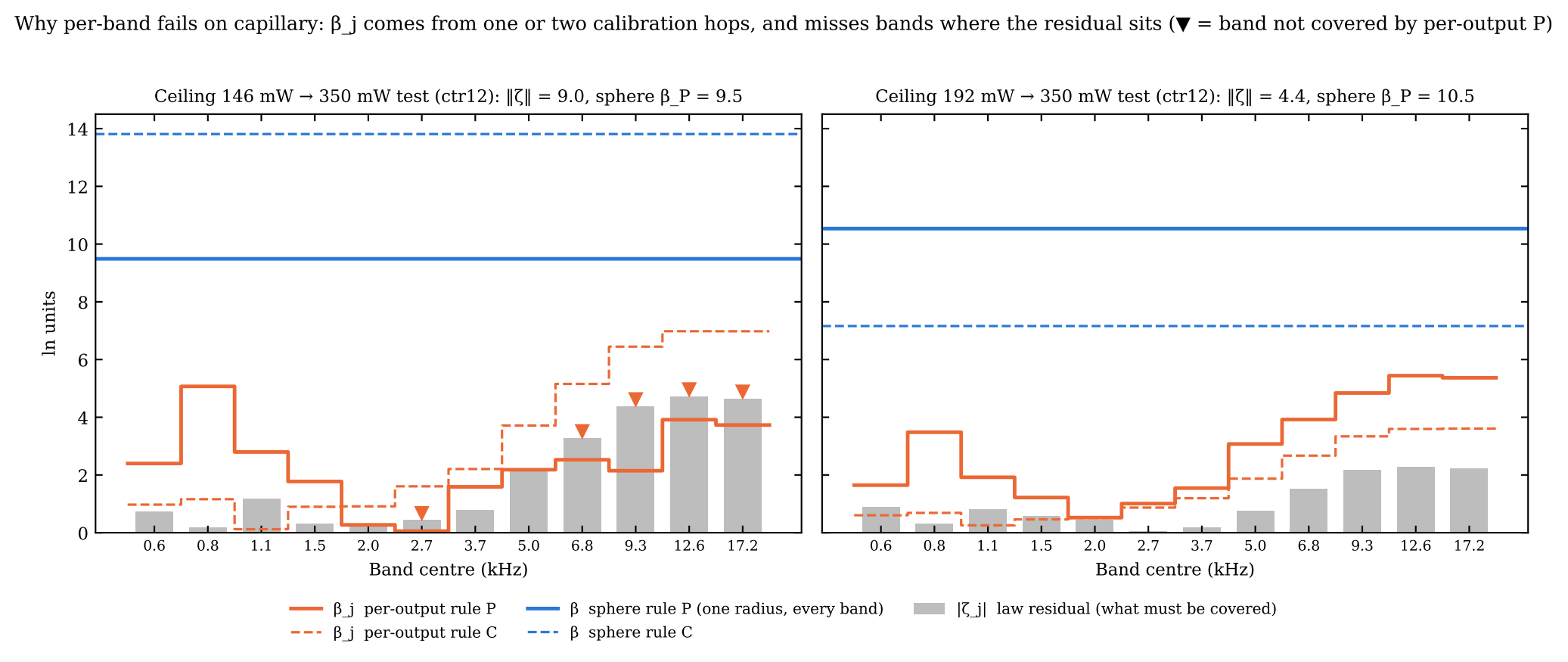

Per-output allowance: the premise breaks more often and S barely moves
Capillary (7 representations)
ceiling 146 mW, rule C
ceiling 146 mW, rule P
ceiling 192 mW, rule C
ceiling 192 mW, rule P
valid 1.00 → 1.00
valid 0.76 → 0.53
valid 1.00 → 1.00
valid 1.00 → 1.00
Wind 2016-18 (development)
below rated, rule C
below rated, rule P
below rated, rule E1
below rated, rule E3
rated, rule C
rated, rule P
rated, rule E1
rated, rule E3
valid 0.22 → 0.19
valid 0.70 → 0.65
valid 0.84 → 0.83
valid 0.97 → 0.96
valid 0.18 → 0.18
valid 0.20 → 0.19
valid 0.20 → 0.19
valid 0.99 → 1.00
Wind 2019 (confirmation, unseen)
below rated, rule C
below rated, rule P
below rated, rule E1
below rated, rule E3
rated, rule C
rated, rule P
rated, rule E1
rated, rule E3
valid 0.22 → 0.19
valid 0.72 → 0.66
valid 0.83 → 0.81
valid 0.94 → 0.93
valid 0.17 → 0.17
valid 0.18 → 0.18
valid 0.18 → 0.18
valid 0.95 → 0.96
0.0 0.2 0.4 0.6 0.8 1.0
0.0 0.2 0.4 0.6 0.8 1.0 1.2 1.4 1.6
Premise held (share of rows)
Median S / e
sphere allowance (current): S = max(D − b − β, 0)
per-output allowance: S_j = max(D_j − b_j − β_j, 0), S = ‖S_j‖


Row by row: the per-output certificate sits on the diagonal (no gain); dotted lines mark S = e
Capillary, rules C and P
median per-output / sphere = 1.03
per-output larger on 92% of rows
premise held (per-output): 35%
Wind 2016-18, rules C, P, E1, E3
median per-output / sphere = 1.01
per-output larger on 70% of rows
premise held (per-output): 33%
Wind 2019 (confirmation)
median per-output / sphere = 1.01
per-output larger on 74% of rows
premise held (per-output): 34%
per-output premise failed
per-output premise held
Per-output S / e
Sphere S / e
Sphere S / e
Sphere S / e


Why per-band fails on capillary: β_j comes from one or two calibration hops, and misses bands where the residual sits (▼ = band not covered by per-output P)
Ceiling 146 mW → 350 mW test (ctr12): ‖ζ‖ = 9.0, sphere β_P = 9.5
Ceiling 192 mW → 350 mW test (ctr12): ‖ζ‖ = 4.4, sphere β_P = 10.5
ln units
0
2
4
6
8
10
12
14
0.6
0.8
1.1
1.5
2.0
2.7
3.7
5.0
6.8
9.3
12.6
17.2
Band centre (kHz)
β_j per-output rule P
β_j per-output rule C
β sphere rule P (one radius, every band)
β sphere rule C
|ζ_j| law residual (what must be covered)